ZIMBABWE SCHOOL EXAMINATIONS COUNCIL (style)
ACADEX PRACTICE EXAMINATION
O-LEVEL · MATHEMATICS

MATHEMATICS 4004/1
PAPER 1 June 2024
2 hours 30 minutes
PAPER 1 — SHORT-ANSWER — CALCULATORS MUST NOT BE USED

These are **original ACADEX questions** written to match ZIMSEC syllabus structure, mark allocations and command words. They are **not** leaked or copied official papers. This script is unique to June 2024 4004/1.

Additional materials: Electronic calculators must NOT be used. Geometrical instruments may be used.
INSTRUCTIONS: Answer all 30 questions. Show all working. The number of marks is given in brackets [ ].

1. In a class of Gifford High, n(A) = 15 (play soccer), n(A ∩ B) = 2 (play both). How many play soccer only? **[2]**

2. Work out 3/7 ÷ 2/2. **[3]**

3. Share $33 in the ratio 4 : 7. **[3]**

4. Find the HCF and LCM of 36 and 42. **[3]**

5. Expand and simplify 5(x + 1) − 5x. **[3]**

6. Simplify 2^5 × 2^3. Leave as a power of 2. **[3]**

7. Work out (5 × 10^6) ÷ 10. Give the answer in standard form. **[3]**

8. Overnight in Harare the temperature was 4°C. By 14:00 it was 22°C. Find the rise. **[3]**

9. Express 90 as a product of its prime factors. **[4]**

10. Find 40% of 250. **[3]**

11. Find the midpoint of (6, 1) and (9, 5). **[3]**

12. Factorise $x^2 - 9$. **[4]**

13. Work out (2 2 ; 3 2)(5 ; 3). **[4]**

14. Make h the subject of $A = 2\pi r(r + h)$. **[4]**

15. A bag contains 3 red, 2 blue and 4 green beads. P(not green)? **[3]**

16. Find the mean of 12, 11, 8, 4, 6. **[3]**

17. Area of a circle radius 14 cm. Take $\pi = 22/7$. **[4]**

18. A cuboid water tank measures 12 m by 3 m by 6 m. Find its volume. **[3]**

19. A regular polygon has 9 sides. Find each exterior angle. **[3]**

20. Solve $4x + 5 = 37$. **[3]**

21. In triangle ABC, $\hat{A} = 66°$ and $\hat{B} = 36°$. Find $\hat{C}$. **[3]**

22. A sequence is 9, 11, 13, 15, … Find an expression for the nth term. **[4]**

23. A right-angled triangle has shorter sides 16 cm and 30 cm. Find the hypotenuse. **[4]**

24. Factorise $x^2 + 7x + 10$. **[4]**

25. In right-angled triangle PQR, $\hat{P} = 30°$ and hypotenuse PR = 8 cm. Find PQ, opposite 30°. **[4]**

26. A trader at Mbare Musika buys a box of tomatoes for $25 and sells for $45. Percentage profit? **[4]**

27. Solve<<
>>x + y = 4<<
>>x − y = 0. **[4]**

28. Write the equation of the line with gradient 2 passing through (0, -1). **[3]**

29. A triangular plot has base 12 m and perpendicular height 7 m. Find its area. **[3]**

30. Solve $3x + 5 < 23$. **[3]**

END OF QUESTION PAPER

Worked solutions and mark-scheme notes are inside the ACADEX app (Extract & Study).